Observe that $J_{2d} \oplus J_{4d} = P^T J_{6d} P$, where $P \in M_{6d}(\mathbb{R})$ is the permutation matrix given by

$$P = \begin{bmatrix} I_d & 0 & 0 & 0 \\ 0 & 0 & I_d & 0 \\ 0 & I_{2d} & 0 & 0 \\ 0 & 0 & 0 & I_{2d} \end{bmatrix}.$$

Then, for all $1 \leq i, k \leq d$ we have

$$\begin{aligned} (P\mathbf{u}_i)^T J_{6d} (P\mathbf{u}_k) &= \mathbf{u}_i^T (J_{2d} \oplus J_{4d}) \mathbf{u}_k = 0 \\ (P\mathbf{v}_i)^T J_{6d} (P\mathbf{v}_k) &= \mathbf{v}_i^T (J_{2d} \oplus J_{4d}) \mathbf{v}_k = 0 \\ (P\mathbf{u}_i)^T J_{6d} (P\mathbf{v}_k) &= \mathbf{u}_i^T (J_{2d} \oplus J_{4d}) \mathbf{v}_k = \delta_{ik}. \end{aligned}$$

Hence, by [Gos06, Theorem 1.15] and (2), there exist vectors $\{\mathbf{u}'_j\}_{j=d+1}^{3d}, \{\mathbf{v}'_j\}_{j=d+1}^{3d} \subseteq \mathbb{R}^{6d}$ such that

$$M := \begin{bmatrix} P\mathbf{u}_1 & \cdots & P\mathbf{u}_d & \mathbf{u}'_{d+1} & \cdots & \mathbf{u}'_{3d} & P\mathbf{v}_1 & \cdots & P\mathbf{v}_d & \mathbf{v}'_{d+1} & \cdots & \mathbf{v}'_{3d} \end{bmatrix} \in M_{6d}(\mathbb{R})$$

is a symplectic matrix. Consequently, $L := P^T M P$ is a $(J_{2d} \oplus J_{4d})$-symplectic matrix. But

$$\begin{aligned} P^T M P &= \begin{bmatrix} \mathbf{u}_1 & \cdots & \mathbf{u}_d & P^T\mathbf{u}'_{d+1} & \cdots & P^T\mathbf{u}'_{3d} & \mathbf{v}_1 & \cdots & \mathbf{v}_d & P^T\mathbf{v}'_{d+1} & \cdots & P^T\mathbf{v}'_{3d} \end{bmatrix} P \\ &= \begin{bmatrix} \mathbf{u}_1 & \cdots & \mathbf{u}_d & \mathbf{v}_1 & \cdots & \mathbf{v}_d & * & * & \cdots & * \end{bmatrix} \\ &= \begin{bmatrix} X & * \\ L_{21} & * \end{bmatrix}, \end{aligned}$$

where '$*$' denotes unspecified block matrices.

$(ii) \iff (iii)$ Assume that $Z := Y + i(J_{2d} - X^T J_{2d} X) \geq 0$. Then we have that $Y \geq 0$ as noticed above. Since $J^T = -J$, the transpose matrix $Z^T = Y - i(J_{2d} - X^T J_{2d} X) \geq 0$. Conversely, the reverse direction follows in a similar manner. This completes the proof. □

Now we are ready to prove our main theorem.

Theorem 3.1. *Let $\Psi : \mathcal{T}(\Gamma(\mathbb{C}^d)) \to \mathcal{T}(\Gamma(\mathbb{C}^d))$ be a quantum channel. Then the following are equivalent:*

(i) The channel Ψ maps Gaussian states to Gaussian states;

(ii) There exist $X, Y \in M_{2d}(\mathbb{R})$ and $\mathbf{w} \in \mathbb{R}^{2d}$ such that for every Gaussian state $\rho_{(\mathbf{m},S)}$, the transformed state $\Psi(\rho_{(\mathbf{m},S)})$ is also a Gaussian state with mean $X^T\mathbf{m} + \mathbf{w}$ and covariance $X^T S X + Y$, i.e.,

$$\Psi(\rho_{(\mathbf{m},S)}) = \rho_{(X^T\mathbf{m}+\mathbf{w}, X^T S X + Y)}; \tag{8}$$

(iii) There exist $X, Y \in M_{2d}(\mathbb{R})$ and $\mathbf{w} \in \mathbb{R}^{2d}$ such that the dual map $\Psi^* : \mathcal{B}(\Gamma(\mathbb{C}^d)) \to \mathcal{B}(\Gamma(\mathbb{C}^d))$ satisfies

$$\Psi^*(W(\mathbf{z})) = \exp\left\{-i\mathbf{w}^T\mathbf{z} - \frac{1}{2}\mathbf{z}^T Y \mathbf{z}\right\} W(X\mathbf{z}), \quad \forall\, \mathbf{z} \in \mathbb{R}^{2d}; \tag{9}$$

(iv) There exists a Gaussian unitary operator U on $\Gamma(\mathbb{C}^{d+d'}) = \Gamma(\mathbb{C}^d) \otimes \Gamma(\mathbb{C}^{d'})$, for some $d' \geq 1$, such that

$$\Psi(\varrho) = \mathrm{tr}_2 \left(U(\varrho \otimes \rho_{(0, I_{2d'})}) U^\dagger \right), \qquad \forall\, \varrho \in \mathcal{T}(\Gamma(\mathbb{C}^d)); \tag{10}$$

(v) The channel $id_{\mathcal{T}(\Gamma(\mathbb{C}^k))} \otimes \Psi$ maps Gaussian states on $\Gamma(\mathbb{C}^k \oplus \mathbb{C}^d)$ to Gaussian states for all $k \in \mathbb{N} \cup \{0\}$. (Here $\Gamma(\mathbb{C}^0) := \mathbb{C}$.)